CPSC 481 - Project 2 Report

Main model: Google Gemini Flash

1. Why did you choose the particular LLM? (2 sentences)

I went with Gemini Flash because it's completely free through Google AI Studio and hooked right into our OpenAI-style setup without any extra headache. Plus, it answered super fast and gave solid, accurate diagnoses across my tests.

2. The major challenge in completing your project (4 sentences)

Three main challenges:

Prompt tuning & format: It was tricky to write a prompt that explained the full Asia network rules clearly, handled missing symptoms, and forced the model to return clean, parseable JSON every single time.

NRP timeouts: The school-hosted gpt-oss model gave good answers, but it kept freezing and hitting 180-second timeouts, which totally broke the GUI.

OpenAI billing: Our testing of OpenAI's API was limited because our free trial credits were already exhausted (credit_balance_exhausted).

We therefore ran multiple tests on the two remaining models, Gemini Flash and NRP-hosted gpt-oss, using eight shared input cases to compare diagnoses and probabilities with our Project 1 baseline and record response times and timeouts.

3. Something surprising you learned about LLMs from this project (3 sentences)

I was honestly surprised that an LLM could pull off exact Bayesian math just from plain-text probability tables. But I quickly realized that getting clean JSON doesn’t mean the numbers inside are actually right. That's why double-checking the model's outputs against our Project 1 baseline is still super important.